

Quality of Schooling, Returns to Schooling and the 1981 Vouchers Reform in Chile

HARRY A. PATRINOS
World Bank, Washington, DC, USA

and

CHRIS SAKELLARIOU*
Nanyang Technological University, Singapore

Summary. — In 1981, Chile introduced nationwide school choice by providing vouchers to any student wishing to attend a “voucher school”. We use a binary instrument based on the reform and unique information on individual cognitive skills to examine the importance of cognitive skills on labor market outcomes. The results suggest that the main beneficiaries of the reform were those who at the time were pupils in basic schooling. Once the treated group is expanded to include older (secondary school age) students, schooling premiums decrease dramatically while the return to cognitive skills increases accordingly, suggesting that a large part of the estimated return from a typical earnings function is due to classical ability bias. Overall, the findings point to heterogeneous effects of the reform.
© 2011 Elsevier Ltd. All rights reserved.

Key words — vouchers, education returns, cognitive skills, instrumental variables

1. INTRODUCTION

When estimating returns¹ to schooling using a Mincer-type earnings function, the disturbance term will capture individual unobservable attributes and effects which, in general, tend to influence the schooling decision, hence resulting in a correlation between schooling and the error term. Cognitive (as well as non-cognitive) ability is such an unobservable; if schooling is endogenous then estimation by ordinary least Squares (OLS) will yield biased estimates of the return to schooling.

The return to investing in education, based on past empirical studies, is known to differ among individuals with different cognitive skills. For example, evidence from the United States (Ingram & Neumann, 2005) shows that over the past decades, individuals with college education but without specific skills experienced the lowest benefits from investing in education. Therefore, individuals with low ability may not benefit as much from investing in education, compared to individuals in the upper part of the ability distribution; for the latter, ability is expected to interact positively with education resulting in higher benefits from education investments.

When a true measure of ability is an omitted variable in the earnings equation, one of three different approaches has been used in the empirical literature. The first uses twins, to arrive at a measure of the causal return to education (see for example, Ashenfelter and Rouse (1998, 1999)). The second approach uses achievement test scores measuring cognitive ability and employs them as additional controls in the earnings function. The third approach uses sources for exogenous variation in educational attainment, such as institutional changes in the schooling system in the form of changes in compulsory schooling laws, abolition of fees, etc, as well as other “natural variations” (i.e., school construction projects) affecting the schooling decision, to estimate a causal return to the education effect using instrumental variable estimation. A combination of the last two approaches is used in this paper.

In this paper, after establishing that the 1981 vouchers reform in Chile increased educational attainment for cohorts around and immediately after the reform, we use cognitive scores as a proxy for education quality and a binary “reform” instrument to investigate the heterogeneous effects of the reform on returns to education and cognitive skills for students in the early stages of education (of primary school age), versus older students who obtained primary school education before the reform. The hypothesis is that children who were not yet of schooling age or pupils at the early stages of schooling benefited more compared to those who received early education before the reform.

We use the International Adult Literacy data (IALS), which have been used in several studies, but not in the context of Chile, the only country in the survey outside Europe and North America. Some of the existing studies using these data (examining a variety of issues, but mostly examining the extent to which the coefficient of the schooling variable in a Mincerian earnings function is affected when cognitive skills are controlled for), include those by Blau and Khan (2001), Leuven, Oosterbeek, and van Ophem (2004), Green and Riddell (2002), and Devroye and Freeman (2001).

2. THE 1981 CAPITATION GRANT (VOUCHERS) SCHEME

Chile introduced a nation-wide school voucher program in 1981, as part of the government’s market-oriented reforms. Prior to the reform there were three types of schools in Chile. Public schools were controlled by the national Ministry of

* We would like to thank anonymous referees for valuable comments which significantly improved this paper. Final revision accepted: March 1, 2011.

Education, which was responsible for all aspects of their operation, including the hiring and payment of teachers, facilities, and curriculum. In 1981, 80 percent of all students were enrolled in public schools. Unsubsidized private schools received no public funding and charged relatively high tuition and catered primarily to upper income households. They accounted for 6 percent of enrollment. Subsidized private schools did not charge tuition and most were religious. Subsidies made up about 50 percent of their running costs. These schools accounted for 15 percent of enrollment.

The 1981 reforms created a nationwide voucher program with incentives for both public and private institutions. The reform included the decentralization of the education system, with public schools being transferred from the Ministry of Education to the municipalities; they became known as municipal schools. Municipal schools would be funded centrally, but each municipality started to receive a per-student payment for every child attending its schools.

Thus, enrollment losses came to have a direct financial effect on the budget.² The reform introduced public funding for private schools. In effect, non-tuition-charging subsidized private schools began to receive exactly the same per-student payment as the municipal schools. These schools are generally known as voucher private schools (Hsieh & Urquiola 2006). Tuition-charging private schools continued to operate without public funding.

Most essential features of this system have remained in place over the last 28 years, although some features, notably the value of the voucher have changed over time (Gallego & Hernando, 2008). Therefore, Chile, among developing countries, is a pioneer and has gone farther than any other country to redefine the roles of the state and private sector in education, by separating the financing from the provision of education.

Soon after the reform, the dropout rate declined sharply (from 8 percent in 1981 to 2.7 percent in 1982 for basic education and from 8.3 percent to 6.2 percent for secondary education). Chile was able to absorb the pressure of secondary school expansion. There was a large shift of students to the private subsidized schools (mainly in major urban areas), whose enrollment increased by 93 percent between 1980 and 1985, at the expense of municipal schools. Despite a decline in public spending for education (from 4 percentage of GDP in 1981 to 2.6 percent in 1990), student intake increased by 42 percent in higher education and by about 15 percentage points in secondary education (Delannoy, 2000). Net enrollment rates in primary schools before the 1981 reform were well below 100 percent; several years later were close to 100 percent, suggestive of the potential effects of the reform on school attainment.

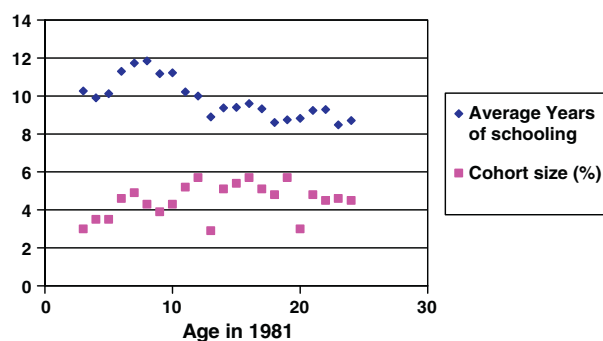


Chart 1. Average years of schooling by cohort and cohort size by age in 1981.

Right after the sweeping 1981 reform, the average years of schooling peak for the cohorts around and immediately after the reform (see Chart 1). Table 3 in the Appendix presents regressions using different combinations of the potential determinants of schooling attainment available in the dataset for individuals aged 22 to 45 (6 to 30 in 1981), in addition to the binary instrument: “of primary school age in 1981”. About 35 percent belong to the primary school age group. The effect of the primary school age dummy on years of schooling consistently suggests an increase of about 1.5 years, a sizable and economically relevant increase.³ It has been suggested (see Bound & Turner, 2007) that if the supply of educational resources, such as expenditure per student, schools, etc. is inelastic, average educational attainment of larger cohorts can be lower than that of smaller cohorts because larger cohorts can “crowd out” educational resources. We, therefore, check whether being born near the voucher reform (before or after), besides being correlated with educational attainment is also correlated with cohort size. Looking at Chart 1, this doesn’t seem to be the case.

Overall, the efficiency of the education system improved after the reform. The number of students educated per unit of public spending increased significantly over the 1980s, while (possibly due to the different methodologies used and inherent problems in methodology) there is no conclusive evidence in the literature that average quality changed significantly. However, improvements in efficiency were achieved at the expense of equity, as subsidized private schools serving a better-informed, better-off population benefited from the opportunities offered by the capitation grants system at the expense of municipal schools which served the less well-off section of the population.

The Chilean voucher system has been studied extensively. Some find that the voucher system had positive impacts on test scores and pre-college examinations (Gallego, 2002, 2004; Contreras and Macias, 2002; Sapelli & Vial, 2002; Sapelli, 2003; Gallego, 2004). Yet others found that there was no impact on test scores, repetition rates, or secondary school enrollment rates (Carnoy & McEwan, 2000; Hsieh & Urquiola, 2006). In addition, Gauri (1998) found that school choice had led to increased social and academic stratification. McEwan et al. (2008) show that segregation did not occur across the board but only in districts large enough to sustain school competition. Bellei (2005) outlines three principal reasons why it is difficult to make comparisons between public and private schools in Chile and how they explain the widely diverging results in individual analyses, all stemming from the lack of random assignment of students to schools: (i) private schools tend to be located in urban areas and serve middle to middle-high income students, (ii) there are wide differences in the level of resources available to schools, even among the same types of schools,⁴ and (iii) there is very little information about how families select schools and how private schools select students.⁵ Thus, it is difficult to measure the supply of private schools, control for school resources, and the estimates are riddled with selection bias. Studies also differ in the ways that they use control variables such as parental education, school socio-economic status, student characteristics, test-score variation, and so on. Gallego (2006) explains that the differences in results can be attributed to changes in the voucher and education systems in the mid-1990s. Hoxby (2003) reiterates that existing studies lead to inconclusive evidence of impact due to a lack of random assignment, thus making it difficult to determine whether variation in school choice is endogenous, and lack of pre-treatment data. This paper does not attempt to directly estimate the impact of the reform on test scores; rather,

assuming that cognitive scores to a large extent reflect education quality, we use a binary “reform” instrument to investigate the heterogeneous effects of the reform on education and cognitive skill premiums for students in the early stages of education versus older students who obtained primary school education before the reform.

3. DATA

The International Adult Literacy Survey (IALS) was carried out in 20 countries between 1994 and 1998, a project undertaken by the governments of the countries and three intergovernmental organizations. It is a carefully designed, innovative survey of adult populations, and goes beyond just measuring literacy capabilities to assessing how these capabilities are applied to everyday activities. Three scales are used to measure literacy: prose literacy, document literacy and quantitative literacy. The questionnaire also included questions about labor market status, earnings, education as well as demographic characteristics.

For Chile, the survey is representative of 98 percent of the population between the ages of 16 and 65 (it excludes residents of institutions and remote areas) and the total number of respondents was 3,583. A four-stage stratified sample was used and stratification was performed according to region and type (urban/rural). As is the case for the rest of the IALS country data, in the Chilean data the three skills are very highly correlated. Consequently, the average of the three scores will be used in the analysis as an aggregate IALS score measure (see also Blau & Khan, 2001; Devroye and Freeman, 2001; Leuven et al., 2004).

4. METHODOLOGY

In the basic Mincerian human-capital model (Mincer, 1974), schooling is assumed to be independent of ability/skills, and that the return from schooling investments is equal for all individuals. However, in the contemporary literature it is acknowledged that the return to schooling must be different for different skill levels. Intuitively, an estimate of the average return to schooling probably over-estimates the return for low skill individuals and under-estimates the return to high skill individuals. One should, therefore, allow the skill level to affect the rate of return to schooling investments.

We incorporate the literacy score as a measure of cognitive skills to proxy for unobserved effects. We expect that the inclusion of a direct measure of cognitive skill will reduce the estimated schooling coefficient (more so for the less skilled), so that the coefficient on education then captures the effect of education alone having controlled for cognitive skills.

The effect of introducing ability/skills differences is two-pronged. First, the more able individuals may be able to ‘convert’ schooling into human capital more efficiently than the less able, and this raises the return to schooling for the more able. In this case, one can conclude that ability and education are complementing each other in producing human capital. On the other hand, the more able may have higher opportunity costs since they may have been able to earn more in the labor market, if ability to progress in school is positively correlated with the ability to earn, and this reduces the rate of return to schooling (Harmon & Walker, 2000).

The model in which both the measure of ability and its interaction with schooling affect earnings and the ability specific re-

turn to schooling is outlined below (Griliches, 1977; Nordin, 2005):

$$\ln w_i = \alpha + \beta S_i + \gamma A_i + \epsilon_i, \quad (1)$$

where w is the hourly wage rate, S years of schooling completed, A is ability and ϵ is an independently distributed error term. Allowing the return to schooling to depend on ability:

$$\ln w_i = \alpha + \beta(f(A_i))S_i + \gamma A_i + \epsilon_i. \quad (2)$$

Using the test score as a proxy for cognitive ability:

$$\ln w_i = \alpha + \beta(f(T_i))S_i + \gamma T_i + \epsilon_i. \quad (3)$$

where T is the IALS test score which is assumed to perfectly measure cognitive ability.⁶ Assuming a linear relationship between test score and cognitive ability:

$$f(T_i) = t_0 + t_1 T_i, \quad (4)$$

we obtain the following earnings equation:

$$\ln w_i = \alpha + t_0 S_i + t_1 T_i S_i + \gamma T_i + \epsilon_i. \quad (5)$$

Finally, including experience, its square, and other covariates:

$$\ln w_i = \alpha + t_0 S_i + t_1 T_i S_i + \gamma T_i + \delta X_i + \epsilon_i. \quad (6)$$

where X is a vector of other covariates.

The coefficient t_0 could be interpreted as the return to schooling for an individual of mean ability.⁷ The coefficient γ measures the approximate percentage change in the hourly wage arising from one standard deviation increase in the score for an individual with no schooling.⁸

Equation (6) is estimated using both Ordinary Least Squares (OLS) to obtain estimates of average return to schooling, as well as IV to estimate the causal effect of education and skills applicable to a group of compliers associated with the 1981 reform. There is a long literature on returns to schooling in Chile (for recent evidence, see for example Sapelli, 2009). The returns to schooling in Chile have been relatively high. In 1978, Corbo and Stelcner (1983) estimated the returns to a year of schooling at 14.8 percent. Between 1960 and 1985, the returns to schooling fluctuated, but increased overall from 11.2 to 15.1 in greater Santiago (Riveros, 1990). In 1987, the returns were 13.7 percent for males and 12.6 percent for females (Psacharopoulos, 1994). By 1989, the returns to an additional year of schooling were still a healthy 12.0 percent (Psacharopoulos & Patrinos, 2004).

5. RESULTS

The group affected by the reform, while heterogeneous, is expected to contain a large proportion of students from higher socio-economic backgrounds and from urban areas. This is because, firstly, in rural areas in Chile, low population density does not permit a choice of schools; and secondly, there is evidence that private-subsidized schools tend to select students to a larger extent than municipal schools (Delannoy, 2000), therefore, better private-subsidized schools facing excess demand practiced screening. According to Gauri (1998), 28 percent of students in the subsidized sector of Santiago had taken tests to be admitted in their current schools. Furthermore, top-up in fees (after 1994), lack of transparency in enrollment procedures, and the cost of uniforms constituted de-facto screening devices. As a result, students who attend private-subsidized schools come from higher-income and better educated families compared to municipal schools (but not private non-subsidized schools).

We estimate the returns to schooling for those induced to make schooling decisions (involving quantity or quality) by the change associated with our instrument—the 1981 reform. In this study, the instrument corresponds to a policy of capitation grants which induced a group of individuals to opt for the opportunities that the policy made available. The estimates, therefore, identify a policy relevant return to schooling in Chile.

A note on identification is due before continuing with the discussion. Identification is based on variations across cohorts as in several other empirical investigations. One such study for Chile is by Bravo, Mukhopadhyay, and Todd (2010), who developed and estimated a model of school attendance and works decisions using 2002 and 2004 panel data. They identify the impacts of the reform from differences in schooling and work choices made and earnings returns received by individuals of similar age who were differentially exposed to the voucher system. They find that the voucher reform significantly increased the demand for private subsidized schools and decreased the demand for both public and non-subsidized private schools. They also find that schooling attainment (especially high school graduation rates) increased, and this applied to individuals from both poor and non-poor backgrounds.

Unlike some studies (for example those by Oreopoulos (2006) and Duflo (2001), which used a unique set of data or a suitable panel), in our case we only have one cross-section. Furthermore, ours being a specialized dataset, we lack certain variables which may have allowed for an alternative identification strategy, such as information at the community level. Not being able to identify (unrestricted) age effects separately from cohort effects, as they are collinear in one cross section, one concern that might arise is that in reality we are using individuals at different points of the life cycle to identify different cohorts and hence different ages; age earnings profiles might vary across age differently by education. This might potentially confound the effect of the instrument with such different profiles. However, we have experience in the regressions, parametrically restricted (i.e., quadratic) to still leave room for identification. It is worth emphasizing that we are conditioning on experience profiles and so we identify using variations across cohorts around these experience profiles. It is also worth mentioning that, unlike most other interventions in the education market used in the literature for IV estimation of returns to schooling which resulted in rather weak instruments, in our case we use a sweeping education reform which results in a binary instrument which is strong enough to allow estimation of IV coefficients with precision.

What is the prior expectation of the magnitude of estimates and in general the nature of the results? Instrumental variable estimates of the return to schooling are expected to differ from OLS estimates. Contrary to expectations, empirical evidence from using a variety of mainly binary instruments, several of which are based on education policy reforms suggest that IV estimates are generally higher than OLS estimates. The dominant explanation for this (as given by Card (2001)) is that because of heterogeneity, there is a distribution of returns and OLS and IV estimates correspond to different weighted averages of this distribution; therefore, IV estimates can exceed OLS estimates. Therefore, the fact that the IV estimates are higher than the OLS estimates is interpreted as an indication that the return to the marginal person (the “switchers”) is higher than that of the average person. Furthermore, Carneiro, Heckman, and Vytlačil (2005) show that the marginal person can have a return that is substantially lower than the return to the average person, and still the

return estimate from IV is greater than the corresponding return from OLS.

Assuming that cognitive skills are associated with quality of schooling, and a group of students switched to better quality schools as a result of the reform, we would expect the affected group (especially those who switching at a younger age) to have higher and less dispersed cognitive skills. Using the IALS dataset, we found that in some countries (such as Scandinavian countries) high cognitive scores are accompanied with low dispersion of scores, and the increase in earnings for one standard deviation increase in scores is generally low compared to Chile (which has the lowest scores among the IALS group of countries—see Table 4 in the appendix). Therefore, the effect of cognitive scores in the IV regressions is expected to be lower compared to the OLS regressions.

The results from OLS regressions for returns to an additional year of schooling and the effects of controlling for the measure of cognitive skills, using a sample of 18–45 years old group of males working for wages, are presented in Table 5. The dependent variable is the logarithm of hourly wage. Inclusion of the direct measure of cognitive ability reduces the return to schooling by about 35 percent, while one standard deviation increase in the score increases earnings by about 17 percent. On average, the interaction effect of schooling with cognitive skills is positive and nearly significant.

The results from IV regressions using the instrument based on the reform are presented in Table 1(A). The binary instrument initially takes the value of 1 for those who at the time of the reform were 2 years or older (18 years or older in) and up to 13 years of age included (29 years) and 0 otherwise, that is preschoolers and those in the 8 years of basic education. Forty percent of the observations belong to the group affected by the reform. Two alternative sample specifications were used: 18–65 and 18–45 years of age. Quantitatively, in both cases the estimates were very similar. However, for the 18–45 group the results were more precise; therefore, we present these results. In the first stage, the correlation of years of schooling with the instrument is strong (partial R^2 is high), and the under-identification test rejects the null hypothesis, indicating that the instrument is sufficiently relevant to explain the potentially endogenous regressor, while the instrument is uncorrelated with the logarithm of hourly wage.

In the standard Mincerian specification, the IV estimate of the return to schooling is about 25 percent higher than the OLS estimate, and ergogeneity of schooling is rejected only at the 20 percent level. Here one needs to consider the composition of the group affected by the reform. This group (which in this case consists of pupils in basic education), contains those who switched to private subsidized schools. While the question of school choice improved the performance of the median student has not been settled in the literature, it has been argued convincingly that the main effect of unrestricted school choice was an exodus of “middle class” students from public sector schooling (Hsieh & Urquiola, 2006) and the practice of screening by private subsidized schools (competing for better students). Furthermore, it is possible that parents tended to select schools which provided good peer groups. Based on this evidence, we attribute the higher IV estimate of the return to schooling to those in a treated group with such characteristics, rather than the average individual.

In column (2), we control for cognitive skill; if we didn’t, cognitive skill would be part of the error term and the instrument is compromised. The assumption of independence implies that the instrument must be independent of cognitive ability. By controlling for cognitive ability we avoid the

Table 1. *Returns to schooling and skills from IV—male employees 18–45 years*

Variable	(1)	(2)	(3)
<i>(A) Instrument = 1 if age in 1981 between 2 and 13 years</i>			
Years of schooling	0.108 (4.8)	0.081 (2.2)	0.082 (2.0)
Experience	0.009 (0.8)	0.009 (0.8)	0.009 (0.7)
Experience squared	0.0001 (0.3)	0.0001 (0.2)	0.0001 (0.2)
Urban	0.152 (1.7)	0.145 (1.7)	0.168 (1.8)
Father white collar	–	–	0.112 (1.4)
Standardized IALS score	–	0.118 (1.5)	0.095 (1.1)
Constant	5.04 (18.5)	5.30 (13.2)	5.26 (11.5)
Centered R^2	0.152	0.178	0.191
N	640	640	549
<i>First stage</i>			
Partial R^2	0.193	0.102	0.100
Cragg–Donald Wald	151.5	72.1	60.1
F-statistic			
[p-Value]	[0.000]	[0.000]	[0.000]
<i>Under-identification test</i>			
H_0 : Matrix under-identified: Cragg–Donald Wald chi-sq-Statistic	152.6	72.8	60.9
[p-Value]	[0.000]	[0.000]	[0.000]
<i>Endogeneity test</i>			
Durbin–Wu–Hausman statistic	1.83	0.68	0.37
[p-Value]	[0.176]	[0.410]	[0.545]
<i>(B) Instrument = 1 if age in 1981 between 2 and 18 years</i>			
Years of schooling	0.066 (3.2)	0.034 (1.2)	0.038 (1.3)
Experience	0.004 (0.3)	0.006 (0.6)	0.005 (0.4)
Experience squared	–0.0000 (0.1)	–0.0001 (0.2)	–0.0000 (0.0)
Urban	0.254 (2.9)	0.207 (2.5)	0.219 (2.5)
Father white collar	–	–	0.137 (1.7)
Standardized IALS score	–	0.208 (3.3)	0.177 (2.6)
Constant	5.50 (22.0)	5.80 (18.7)	5.73 (16.9)
Centered R^2	0.159	0.182	0.193
N	640	640	549
<i>First stage</i>			
Partial R^2	0.233	0.179	0.192
Cragg–Donald Wald	192.9	137.9	128.6
F-statistic			
[p-Value]	[0.000]	[0.000]	[0.000]
<i>Under-identification test</i>			
H_0 : Matrix under-identified: Cragg–Donald Wald chi-sq-statistic	194.5	139.2	130.3
[p-Value]	[0.000]	[0.000]	[0.000]
<i>Endogeneity test</i>			
Durbin–Wu–Hausman statistic	0.63	0.55	0.57
[p-Value]	[0.429]	[0.458]	[0.450]

Note: Theoretical sampling weights included; *t*-Values in parentheses.

problem of using possibly invalid instruments and make our estimates more credible (Carneiro et al., 2005).

When we include the measure of cognitive skills in the equation, the estimate of the return to schooling decreases by about 16 percent (which is much less than the reduction one sees in

the OLS regressions), while the contribution of one standard deviation in the cognitive score to earnings is one-half of the corresponding estimate from the OLS regressions (8.5 compared to 17 percent). The above results seem to indicate that students who switched to private subsidized schools and in 1981 were of basic school age, were part of a rather homogeneous group of students with above average cognitive skills. That is, “better” students formed the bulk of those who switched from public to private subsidized schools. As a result, only a small part of the return to schooling estimate is due to classical ability bias. Column (3) indicates that the contribution of cognitive skills is mainly through their interaction with schooling (as was the case in the OLS regressions).

Another instrument (in addition to the policy reform based instrument) is included in the specification presented in Table 6, namely father’s years of schooling.⁹ Here the objective is twofold: first, test the over-identifying restrictions and, second, test the hypothesis that if the treated group is an even more privileged one (have taken advantage of the capitation grant system and have more educated fathers), its cognitive skills will be better and more homogeneous compared to the treated group in Table 1(A) (which in turn had better and more homogeneous cognitive skills compared to the average individual). Therefore, the contribution of cognitive skills to earnings should be small, while returns to schooling should increase.

Based on the value of Hansen’s J-statistic (see Hansen, 1982), the null hypothesis (that the instruments are appropriately uncorrelated with the disturbance process) is easily accepted in all cases. As hypothesized, the contribution of cognitive skills to earnings is nearly zero. At the same time, the estimate of the return to schooling increases by between 15–20 percent compared to Table 1(A).

In Table 1(A) the binary instrument is modified and takes the value of 1 for those who at the time of the reform were 2 years or older and up to 18 years of age included and 0 otherwise; that is the group includes all those who could have been affected by the reform, either by switching to private subsidized schools because of their perceived higher quality compared to municipal schools, or by choosing to continue schooling at the secondary level as a result of the increase in the supply of private schools associated with the education reform, which led to a large increase in secondary school participation. Estimates of the return to schooling are now lower nearly 40 percent in the Mincerian specification and by more than 50 percent when we control for cognitive skills. At the same time, one standard deviation increase in the cognitive score increases earnings by over 20 percent, up from just over 10 percent in Table 1(A). The coefficient estimates are now close to those from OLS; as a result the endogeneity test easily accepts the null hypothesis that the difference in coefficients between IV and OLS are not systematic.

In Table 7 in the appendix we use the expanded age group (ages 2–18 in 1981) with the additional instrument (father’s years of schooling). Here again, the effect of including the secondary school age students results in lower estimates for the return to schooling and higher estimates for the return to cognitive score compared to the results in Table 6. However, the return to schooling estimate is still higher and the return to cognitive score is lower than in Table 1(B), given the composition of the treated group which consists of students with educated fathers. Once again, based on the value of Hansen’s J-statistic, the null hypothesis is accepted.

Finally, we ask the question: what happens if we progressively expand the treated group to include students who, at the time of the reform were between the ages of 4 (20 years old in 1997) and 20 (36 years old in 1997)? That is, starting

with a treated group of very young children in 1981, and progressively expanding the group one year at a time until age 20, tracing the changes coefficient estimates for schooling and cognitive skills. Suppose that such students, before deciding whether to switch schools (based on the comparison between expected benefits and costs) were enrolled in, say, municipal schools. Assuming that municipal schools on average provide lower quality schooling, switching to private-subsidized schools at a younger age (such as the beginning of primary school) should result in a higher and more homogeneous cognitive skill endowment by the end of the schooling period, compared to switching at a later age (such as during high school). We, therefore, would expect that as we progressively increase the instrument upper cutoff age, the treated group becomes less endowed in cognitive skills (because they have received most of their schooling before 1981) and less homogeneous in these skills. As a result, as we add the marginal individual in the treated group, the return to cognitive skills will be increasing and the return to schooling will be decreasing. This is because part of a given estimate of the return to schooling from the Mincerian specification is due to the independent effect of cognitive skills. Given a group with less and more dispersed cognitive skills, a larger part of the Mincerian estimate of the return to schooling will be due to the independent effect of cognitive skills. The opposite is true for a group endowed with more and less dispersed cognitive skills).

Charts 2 (based on the Mincerian specification) and 3 (addition of score), confirm this hypothesis. Chart 2 also reinforces the earlier finding that attending school after 1981 not only had a positive effect on school attainment, but also the effect seems to be stronger for individuals belonging to cohorts that should have been affected more (primary school children). In Chart 2, the Mincerian return is around 9–12 percent for the cohorts who in 1991 were of primary school age or lower and subsequently (by the end of basic education) declines sharply. In Chart 3, the return to schooling fluctuates at about 7–10 percent for the primary school cohorts and subsequently

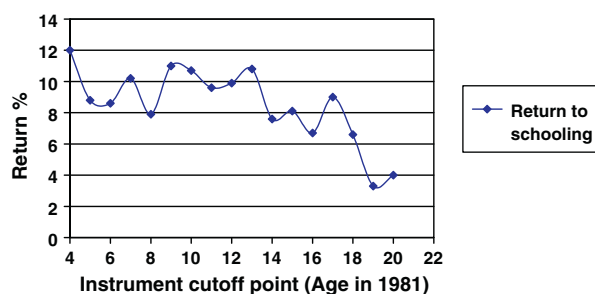


Chart 2. Returns to an additional year of schooling for different instrument cutoff points.

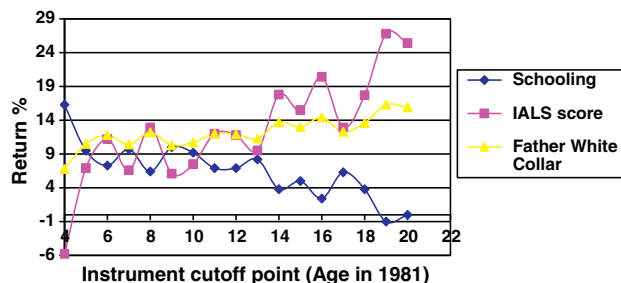


Chart 3. Returns to schooling, cognitive score and having a father in white collar occupation for different instrument cutoff points (years of schooling instrumented).

declines sharply; the opposite pattern is observed for the return to the cognitive score. With the last expansions of the instrument cutoff point, the return to schooling approaches zero, while one standard deviation increase in the cognitive score increases earnings by 20–25 percent. Finally, the independent effect of having a father in a white collar occupation increases slightly along with the increase in the effect of the cognitive score. One can also observe that the decline in the return to schooling estimate (increase in the return to cognitive score) coincides with the period between the end of the two cycles of compulsory basic education in Chile and the end of secondary education.

Before commenting on the above presented results we recognize that, while they are suggestive, the results are not robust to alternative explanations. This is because the use of the policy related binary instrument allows for instrumenting the potentially endogenous years of schooling but not the cognitive score. However, one should expect that cognitive scores were also affected by the reform. Thus, the above approach does not allow for disentanglement of the effect of the reform between school attainment and learning. The fact that scores are not instrumented allows for an alternative statistical explanation, if one assumes that both years of schooling and cognitive scores are measured with (classical) measurement error. Then, in the IV regression, as one variable (years) is projected using the instrument, it is reasonable to expect that the coefficient of the instrumented variable will increase in value while the coefficient of the score variable will decrease. In dealing with this problem one needs additional instruments. The only potential instruments in the data set which are expected to be correlated with both the potentially endogenous regressors are family background variables. However, family background variables, such as father's and mother's education, have been criticized as generally unsuitable because they don't meet the *strict validity* assumption required in IV regressions.

In a recent study, Hoogerheide, Block, and Thurik (2010) used Bayesian analysis to study the degree to which violations of the strong validity assumption from using parental education instruments affect the estimation results. They show that in the case of moderate direct effect of the instrument on the dependent variable (earnings), the results do not deviate much from the benchmark case of perfect instrument validity; that is the size of the bias is small and the violation of the strict validity assumption does not necessarily lead to significantly different results, thus injecting confidence in the use of family background variables as instruments in income regressions. Their analysis considers a small direct effect of the instrument on the income variable, which works in addition to its indirect effect via the individual's own education, controlling for a vector of other characteristics, X_i within the model:

$$\text{Education} = a + bZ + \sum_{i=1}^m C_i X_i + u \quad (7)$$

$$\text{Income} = \alpha + \beta \text{Schooling} + \gamma Z + \sum_{i=1}^m \delta_i X_i + v \quad (8)$$

where Z is the instrument used (such as father's education).

The ratio of the effects of the instrument and education, γ/β , would equal 0 for a perfectly valid instrument. This ratio is likely to be positive (as one would expect the direct effect of one's education and the indirect effect through father's education to be of the same sign) and unlikely to be very large (as we would expect one's own education to be more important than the father's education). The authors show that while relaxing the strict validity assumption does lead to different estimates,

the size of the bias is small for small to moderate values of the ratio γ/β . In fact, the estimates remain qualitatively similar even when the validity of the instrument would be substantially violated, compared to the benchmark assumption of a perfectly exogenous instrument.

In our case, the independent effect of father's years of schooling is small, at about 0.01, suggesting a one percent increase in earnings per additional year of father's education (see Table 8). This effect was statistically significant only at the 20% level. The estimate of the ratio γ/β is approximately 0.17 to 0.18. The corresponding ratio with respect to the cognitive score is much smaller. Replacing father's years of schooling with mother's years, results in a direct effect on earnings which is very small and insignificant.¹⁰ Noting that the intention of instrumenting is not to derive precise estimates of the average return to schooling in Chile, but to qualitatively distinguish the relative effects of quantity of education versus learning in relation to the 1981 reform, we proceed to use the binary variable associated with the reform, along with father's and mother's years of schooling as a combination, to instrument for both years of schooling and the cognitive score. After examining the contribution of each instrument, the binary reform instrument is stronger than the parental education instruments.

Table 2 presents the results corresponding to Table 1(A), but after instrumenting for both potentially endogenous variables. In reporting first stage results, of interest are: the partial R^2 for each potentially endogenous regressor, the partial R^2 of

instrument relevance after taking into account inter-correlations among instruments, as well as test results for under-identification and weak identification. While the partial R^2 is high in all cases, the Shea R^2 for those in the 2–13 age group in 1981 is rather small, but the hypothesis that the equation is under-identified is rejected with p-values of 0.03 to 0.06; the weak identification test, however, shows that the combination of instruments is weak in identifying two potentially endogenous regressors. For the group in the 2–18 age group in 1981, the under-identification hypothesis is easily rejected and the combination of instruments is borderline weak. The statistic for the over-identification test is in the acceptable range in all cases (p -Value for Hansen's J-statistic between 0.2 and 0.33). Endogeneity tests tend to reject endogeneity. Finally, redundancy tests reject the hypothesis that any of the instruments is redundant. From the first stage results, therefore, it is evident that the fact that any parent's years of schooling are relatively strong instruments in instrumenting for only years of schooling, does not mean that the combination of instruments is strong in instrumenting for both the potential endogenous regressors.

Coefficient estimates lack precision due to a generally weak set of instruments, exacerbated by a rather small sample size. In the case of the expanded group, estimates are relatively more precise, but the point estimates of the coefficient of years of schooling are very small. However, the results are qualitatively similar to those in Table 1(A) and (B) and give the same picture: The return to schooling is 2 to 4 times higher for the

Table 2. Returns to schooling and skills from IV—male employees 18–45 years (instruments: Reform + Parents' years of schooling)

Variable	2–13 years old in 1981		2–18 years old in 1981	
	(1)	(2)	(1)	(2)
Years of schooling	0.060 (0.6)	0.076 (0.8)	0.014 (0.3)	0.041 (0.7)
Experience	0.016 (1.2)	0.020 (1.3)	0.013 (1.0)	0.017 (1.2)
Experience squared	−0.0002 (0.6)	−0.0004 (1.0)	−0.0003 (0.7)	−0.0004 (1.0)
Urban	0.106 (1.0)	0.129 (1.2)	0.149 (1.4)	0.169 (1.6)
Father white collar	—	0.101 (1.1)	—	0.112 (1.1)
Standardized IALS score	0.278 (0.7)	0.158 (0.4)	0.413 (1.7)	0.249 (1.0)
Constant	5.50 (5.6)	5.31 (4.8)	5.96 (10.8)	5.68 (10.0)
Centered R^2	0.209	0.228	0.173	0.218
N	512	478	512	478
<i>First Stage</i>				
Partial R^2 for years of Schooling	0.255	0.237	0.304	0.300
Cragg–Donald Wald F-statistic	57.5	48.7	73.7	67.2
[p-value]	[0.000]	[0.000]	[0.000]	[0.000]
Partial R^2 for Cognitive Score	0.193	0.170	0.169	0.140
Cragg–Donald Wald F-statistic	40.3	32.1	34.1	25.4
[p-value]	[0.000]	[0.000]	[0.000]	[0.000]
Shea partial R^2 for years of Schooling	0.020	0.018	0.066	0.068
Shea partial R^2 for Cognitive Score	0.015	0.013	0.036	0.032
<i>Under-identification test</i>				
H_0 : Matrix under-identified: Cragg–Donald Wald chi-sq-statistic	7.0	5.6	18.9	15.5
[p-Value]	[0.030]	[0.061]	[0.000]	[0.000]
<i>Weak-identification test</i>				
H_0 : Equation Weakly Identified: Cragg–Donald	2.30	1.84	6.21	5.07
Wald F-Statistic Critical Values for Stock–Yogo Test:				
10–25% Max. IV Size	13.4–5.4	13.4–5.4	13.4–5.4	13.4–5.4
<i>Over-identification test</i>				
Hansen's J-statistics	1.64	1.23	1.35	0.96
[p-Value]	[0.200]	[0.267]	[0.246]	[0.327]
<i>Endogeneity test</i>				
Durbin–Wu–Hausman statistic	1.73	0.37	1.21	0.045
[p-Value]	[0.421]	[0.831]	[0.546]	[0.800]

Note: Theoretical sampling weights included; t-values in parentheses.

group of primary school age in 1981, while the effect of one standard deviation increase in the score is about 50 to 60 percent smaller. **Chart 4** reproduces the results of **Chart 3**, that is expanding the treated group to include progressively older school age children at the time of the reform. Relying on point estimates which lack precision, the same pattern is observed: as in **Chart 3**, beyond the age of 10 the return to schooling declines and the return to cognitive score shows a corresponding increase. The effect of having a father in a white collar occupation remains stable throughout in **Chart 4**, at about 10 percent, compared to a moderate increasing pattern in **Chart 3**.

One observation worth commenting on is that even within the group of primary education age in 1981, those in school closer to 1981, that is those who attended more years of primary education after the reform are more affected than those of primary school age who attended fewer years after the reform. For example, based on graph 4, children of age 9 or less after the reform are associated with an average return to schooling of 10 percent compared with 6 percent for primary school children aged 10–13. This provides additional evidence of a clear change in attainment around the reform.

The evidence could be taken to suggest that the treated group which benefited most from the 1981 “voucher” reform, besides coming mainly from a better socioeconomic background and from urban areas, were enrolled in the early stages of schooling. They could have benefited from better quality schooling which endowed them with better schooling cognitive skills (compared to the untreated). As a result, given the “abundance” of schooling cognitive skills in this group, the labor market reward for an additional year of schooling reflects the “true” non-cognitive return to schooling.

However, the cohorts that obtained early childhood schooling before 1981 do not seem to have benefited as much. The reform facilitated the absorption of the large secondary school expansion, despite the decline in public spending for education. However, because of the secondary school expansion, lower ability students (from “middle class families or otherwise) may have been admitted into the system, students who had earlier received their basic education in public schools. This group entered the labor market less endowed in schooling cognitive skills. As a result, the labor market rewarded these skills significantly; failure to control for such skills in the earnings function results in a return to schooling estimate which is seriously biased upwards.

Finally, we also tried to explore potential biases related to unobservables using an approach suggested in [Altonji, Elder, and Taber \(2005\)](#). The main part of this approach estimates the bias on “a” in OLS models for:

$$Y = aTR + XG + e$$

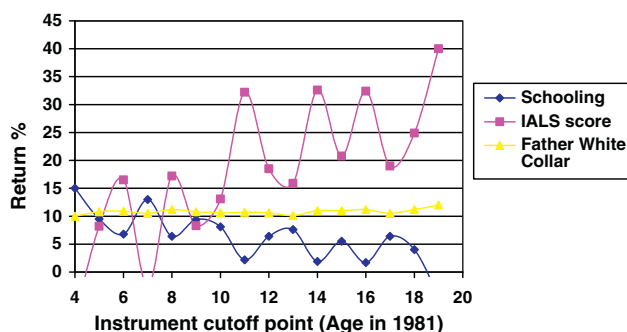


Chart 4. Returns to schooling, cognitive score and having a father in white collar occupation for different instrument cutoff points (years of schooling and score instrumented).

using the condition that: $\text{Cov}(e, TR) / \text{var}(e) = \text{Cov}(XG, TR) / \text{var}(XG)$, where Y is the outcome of interest, in our case earnings, and TR is the potentially endogenous treatment, in our case years of schooling or the cognitive score. Estimates of “G” are obtained under the null hypothesis that $a = 0$. Bias estimates are obtained with analytic standard errors of the bias estimates.

This approach is mostly suited and has been used in the context of binary treatment, such as Catholic school attendance and its causal effects on various outcomes (as in [Altonji et al. 2005](#)), or unionization of teachers and their causal effects on test scores ([Kingdon & Teal 2010](#)). In our case, in place of the binary treatment we have continuous potentially endogenous treatments and our objective is not to derive causal effects but to “tease out” the heterogeneity in the coefficients of the treatments depending on the age of students when the reform took effect. Nevertheless, we derived estimates of the bias on “a” (effect of an additional year of schooling). These estimates varied little and ranged between 0.44 and 0.6 depending on the equation specification. The estimates of standard errors of this bias implied relatively low statistical significance (only nearly significant at the 5% level). Considering the ratio of “a” to the bias estimate (to which [Altonji et al. \(2005\)](#) assign a specific interpretation), using and extended Mincerian earnings function specification, this ratio is about 0.2. According to the interpretation given to this ratio, it could imply that the ratio of selection on unobservables to selection on observables required to explain away all of the effect of an additional year of schooling is not very large.¹¹ Concluding, unobservables may play some role in our estimates, but not enough to wipe away the effect of the treatment variable.

6. CONCLUSIONS

We use the sweeping school reform of 1981 in Chile to identify a binary instrument and estimate returns to schooling from IV, after controlling for cognitive ability. Given that ability bias needs to be dealt with, accounting for cognitive ability avoids the problem of using a possibly invalid instrument.

The 1981 capitation grant scheme facilitated the absorption of the pressure of secondary school expansion which resulted from the capitation grants paid to schools and the opening of the education system to the private sector, and resulted in large increases in student intake as well as a sharp decline in the dropout rate, especially in basic education. Overall, the reform improved the efficiency of the education system, possibly at the expense of equity.

The results suggest that those who in 1981 switched to private subsidized schools were mainly urban “middle class” students leaving public schools, possibly in search of better peer groups. We find evidence suggesting that for students with such characteristics, who switched during their basic education, the labor market reward for an additional year of schooling is a measure of the non-cognitive return to schooling. This is not the case, however, for older cohorts (those in secondary education). The large secondary school expansion seems to have attracted a heterogeneous group of students who had earlier received their basic education in public schools. For this group of students the labor market rewarded these skills significantly, while the non-cognitive component of the return to schooling for this group is rather small.

Overall, the findings show that attending school around and soon after the reform had a positive effect on school attainment. They also suggest that the 1981 reform relaxed certain binding constraints. Finally the finding that the effects of the reform are stronger for those who were of primary school

age in 1981 points to heterogeneous effects of the reform. In previous research policy instruments (especially those relating to education reforms) have been used mainly to correct for endogeneity bias and derive average treatment effects by comparing effect on the treated versus the untreated. Based on the

findings of this paper, another avenue of research could be to investigate heterogeneous effects of reforms within the group of treated and effects on quantity (effect on attainment) versus quality of education.

NOTES

1. We recognize that there is no such thing as the “return to schooling” in a pure sense. The return to schooling is the equilibrium price that results from the interplay of demand and supply (see Goldin & Katz 2008).
2. There is evidence that that this was not the case at least for a subset of public schools (see Sapelli 2003; Gallego 2006; Gallego & Hernando 2008).
3. Bravo et al. (2010) also found that schooling attainment (especially high school graduation rates) increased.
4. Sapelli and Vial (2002) deal with this criticism.
5. However, this criticism is not consistent with information on the implications of selection from the supply side provided in Contreras, Bustos, and Sepulveda (2009) and Gallego and Hernando (2007), Gallego and Hernando (2008); furthermore, Contreras et al. (2009) provide newer and more detailed evidence on the extent of selection on the part of schools.
6. We recognize this as a strong assumption.
7. The test score variable used in the analysis is standardized, with mean 0 and standard deviation 1.
8. For an individual with mean years of schooling, the impact is measured by $\gamma + t_1^*$ (mean years of schooling).
9. The rationale for using instruments based on family background variables is discussed later on in this section.
10. We used nonlinear hypothesis testing to test the significance of the γ/β ratio as a post-estimation F-test. Significance is rejected in both the cases involving father’s and mother’s education with p-values of 0.234 and 0.802 respectively.
11. Estimates of this bias derived from extended Mincerian earnings functions using a variety of large data sets from several countries were of similar magnitude, ranging from 0.15 to 0.5.

REFERENCES

- Altonji, J. G., Elder, T. E., & Taber, C. R. (2005). Selection on observed and unobserved variables: Assessing the effectiveness of catholic schools. *Journal of Political Economy*, 113(1), 151–184.
- Ashenfelter, O., & Rouse, C. E. (1998). Income, schooling and ability, evidence from a new sample of identical twins. *Quarterly Journal of Economics*, 113(1), 253–284.
- Bellei, C. (2005). The Private–Public School Controversy: The Case of Chile. *Conference on Mobilizing the Private Sector for Public Education*, October 5–6, Harvard University.
- Blau, F., & Khan, L. (2001). International differences in male wage inequality: institutions versus market forces. *Journal of Political Economy*, 104, 791–837.
- Bound, J., & Turner, S. E. (2007). Cohort crowding: How resources affect collegiate attainment. *Journal of Public Economics*, 91(5–6), 877–899.
- Bravo, D., Mukhopadhyay, S., & Todd, P. (2010). Effects of school reform on education and labor market performance. Evidence from Chile’s universal voucher system. *Quantitative Economics*, 1(1), 47–95.
- Card, D. (2001). Estimating the return to schooling: progress on some persistent econometric problems. *Econometrica*, 69(5), 1127–1160.
- Carneiro, P., Heckman, J. J., & Vytlacil, E. (2005). *Understanding what instrumental variables estimate: Estimating marginal and average returns to education*. Department of Economics: University of Chicago.
- Carnoy, M., & McEwan, P. J. (2000). The effectiveness and efficiency of private schools in Chile’s voucher system. *Educational Evaluation and Policy Analysis*, 22(3), 213–239.
- Contreras, D., & Macias, V. (2002). *Competencia y resultados educacionales*. Santiago: University of Chile (Processed).
- Contreras, D., Bustos, S., & Sepulveda, P. (2009). When schools are the ones that choose: the effect of screening in Chile. In F. Barrera-Osorio, H. A. Patrinos, & Q. Wodon (Eds.), *Emerging evidence on private participation in education: Vouchers and faith-based providers*. The World Bank: Washington DC.
- Corbo, V., & Stelcner, M. (1983). Earnings determination and labour markets: gran santiago Chile 1978. *Journal of Development Economics*, 12, 251–266.
- Delannoy, F. (2000). *Education reforms in Chile, 1980–1998: A lesson in pragmatism. education reform and management publication series, Vol. 1 No. 1*. Washington, DC: The World Bank.
- Devroye, D. and R. Freeman. (2001). Does inequality in skills explain inequality of earnings across advanced countries? *NBER Working Paper 8140*.
- Duflo, E. (2001). Schooling and labor market consequences of school construction in Indonesia: Evidence from an unusual policy experiment. *American Economic Review*, 91(4), 795–813.
- Gallego, F. A. (2002). Competencia y Resultados Educativos. Teoría y Evidencia para Chile. *Cuadernos de Economía*, 39(118), 309–352.
- Gallego, F. A. (2004). *School choice, incentives, and academic outcomes: evidence from Chile*. Boston: Massachusetts Institute of Technology (Processed).
- Gallego, F. A. (2006). *Voucher-school competition, incentives and outcomes: evidence from Chile*. Boston: Massachusetts Institute of Technology (Processed).
- Gallego, F. A., & Hernando, A. (2007). *School choice in Chile: Looking at the demand side*. Instituto de Economía, Pontificia Universidad Católica de Chile: Documentos de Trabajo, 356.
- Gallego, F. A., & Hernando, A. (2008). On the determinants and implications of school choice. Semi-structural specifications for Chile. *Economía*, 9(1), 197–244.
- Gauri, V. (1998). *School choice in Chile: Two decades of educational reform*. Pittsburgh: Pittsburgh University Press.
- Goldin, C., & Katz, L. (2008). *The race between education and technology*. Cambridge Ma: Harvard University Press.
- Green, D. A., & Riddell, C. (2002). *Literacy and earnings: An investigation of the interaction of cognitive and non-cognitive attributes in earnings generation*. Department of Economics, University of British Columbia, Canada: Discussion Paper: 02–11.
- Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50, 1029–1054.
- Harmon, W. C., & Walker, I. (2000). The returns to quantity and quality of education: Evidence for men in England and Wales. *Economica*, 67(265), 19–35.
- Hoogerheide, L., Block, J. H., & Thurik, R. (2010). *Family background variables as instruments for education in income regressions: A Bayesian analysis*. Amsterdam, The Netherlands: Tinbergen Institute Discussion Paper, No 075/3.

- Hoxby, C. M. (2003). School choice and school competition: Evidence from the united states. *Swedish Economic Policy Review*, 10, 11–67.
- Hsieh, C. T., & Urquiola, M. (2006). The effects of generalized school choice on achievement and stratification: Evidence from Chile's voucher program. *Journal of Public Economics*, 90, 1477–1503.
- Ingram, B., & Neumann, G. (2005). The returns to skill. *Labour Economics*, 13(1), 35–59.
- Kingdon, G., & Teal, F. (2010). Teacher unions, teacher pay and student performance in India: A pupil fixed effects approach. *Journal of Development Economics*, 91, 278–288.
- Leuven, E., Oosterbeek, H., & van Ophem, H. (2004). Explaining international differences in male skill wage differentials by differences in demand and supply of skills. *Economic Journal*, 114, 466–486.
- Oreopoulos, P. (2006). Estimating average and local average treatment effects when compulsory schooling laws really matter. *American Economic Review*, 96(1), 152–175.
- Psacharopoulos, G. (1994). Returns to investment in education: A global update. *World Development*, 22(9), 1325–1343.
- Psacharopoulos, G., & Patrinos, H. A. (2004). Returns to investment in education: A further update. *Education Economics*, 12(2), 111–134.
- Riveros, L. (1990). The economic return to schooling in Chile. An analysis of its long-term fluctuations. *Economics of Education Review*, 9(2), 111–121.
- Sapelli, C., & Vial, B. (2002). The performance of private and public schools in the Chilean voucher system. *Cuadernos de Economía*, 39(118), 423–454.
- Sapelli, C. (2003). The Chilean Voucher System: some new results and research challenges. *Cuadernos de Economía*, 40(121), 530–538.
- Sapelli, C. (2009). *Los Retornos a la Educacion en Chile: Estimaciones por Corte Transversal y por Cohortes*. Catholic University of Chile: Mimeo.

APPENDIX A

See Tables 3–8.

Table 3. *Effect of reform on years of schooling—male employees 18–45 years*

Dependent variable: Years of schooling	(1)	(2)	(3)	(4)
Primary school age in 1981	1.47 (5.6)	1.44 (2.8)	1.47 (3.0)	1.82 (3.5)
Age	—	0.45 (2.5)	0.38 (2.3)	0.42 (2.3)
Age squared	—	−0.007 (2.8)	−0.006 (2.6)	−0.006 (2.1)
Urban	—	—	3.16 (9.8)	2.04 (5.4)
Father's years of schooling	—	—	—	0.279 (7.7)
Father white collar	—	—	—	0.685 (1.9)
Constant	9.71 (55.7)	3.25 (1.0)	1.63 (0.6)	−1.21 (0.4)
Adj. R^2	0.040	0.050	0.162	0.294
N	723	723	723	552

Note: Theoretical sampling weights included. t -Values in parentheses.Table 4. *Country sample means: Years of schooling and IALS score*

Country	Mean years of schooling	IALS/10	Effect of IALS score on earnings (rank)
Canada	13.1 (3.5)	2.92 (0.60)	7
Chile	9.6 (4.0)	2.15 (0.58)	1
Czech Republic	13.0 (2.8)	2.89 (0.47)	11
Denmark	13.2 (3.3)	2.96 (0.40)	13
Finland	12.8 (3.5)	2.99 (0.43)	12
Germany	11.8 (3.5)	2.92 (0.47)	6
Hungary	12.3 (3.1)	2.63 (0.48)	2
Italy	11.1 (4.1)	2.52 (0.62)	13
Netherlands	13.4 (4.2)	2.94 (0.43)	5
Norway	12.1 (2.8)	3.00 (0.41)	10
Poland	11.7 (3.0)	2.43 (0.64)	15
Slovenia	11.5 (3.0)	2.40 (0.61)	9
Sweden	11.8 (3.7)	3.12 (0.51)	8
Switzerland	12.9 (3.4)	2.83 (0.54)	4
United States	13.9 (3.2)	2.85 (0.65)	3

Source: Leuven et al. (2004).

Note: Standard deviations in parentheses.

Table 5. *Returns to schooling from OLS—male employees 18–45 years*

Variable	(1)	(2)	(3)	(4)
Years of schooling	0.081 (8.1)	0.053 (4.5)	0.048 (4.0)	0.061 (4.5)
Experience	0.006 (0.5)	0.008 (0.7)	0.104 (1.2)	0.015 (1.2)
Experience squared	0.0000 (0.0)	−0.0000 (0.0)	−0.0002 (0.7)	−0.0003 (0.7)
Urban	0.219 (2.9)	0.182 (2.4)	0.197 (2.6)	0.143 (1.6)
Standardized score	—	0.172 (4.3)	0.025 (0.3)	0.144 (3.1)
Stand. score–Years of schooling interaction	—	—	0.015 (2.0)	—
Father's years of schooling	—	—	—	0.010 (1.2)
Constant	5.35 (35.2)	5.61 (34.7)	5.58 (34.5)	5.45 (29.5)
Adj. R^2	0.157	0.179	0.183	0.199
N	640	640	640	535

Note: Theoretical sampling weights included; t -Values in parentheses.

Table 6. *Returns to schooling from IV—male employees 18–45 years (instrument = 1 if age in 1981 between 2 and 13 years)*

Variable	IV: Reform + father's years of schooling	IV: Reform + father's years of schooling	IV: Reform + father's years of schooling
Years of schooling	0.128 (4.6)	0.111 (2.9)	0.097 (2.4)
Experience	0.016 (1.0)	0.016 (1.2)	0.021 (1.5)
Experience squared	−0.0001 (0.3)	−0.0001 (0.4)	−0.0003 (0.8)
Urban	0.091 (0.9)	0.099 (1.0)	0.122 (1.2)
Father white collar	—	—	0.111 (1.3)
Standardized score	—	0.060 (0.7)	0.067 (0.8)
Constant	4.82 (14.3)	4.99 (11.9)	5.09 (11.5)
Centered R^2	0.170	0.186	0.205
N	535	535	499
<i>First Stage:</i>			
Partial R^2	0.243	0.125	0.121
Cragg–Donald Wald F-statistic	85.1	37.6	33.7
[p -Value]	[0.000]	[0.000]	[0.000]
<i>Under-identification test</i>			
H_0 : Matrix under-identified:	172.1	76.2	68.6
Cragg–Donald Wald chi-sq-statistic			
[p -Value]	[0.000]	[0.000]	[0.000]
<i>Over-identification test</i>			
Hansen's J-statistic	0.44	0.32	0.08
[p -Value]	[0.508]	[0.574]	[0.781]
<i>Endogeneity test</i>			
Durbin–Wu–Hausman statistic	3.97	1.82	0.64
[p -Value]	[0.046]	[0.177]	[0.424]

Note: Theoretical sampling weights included. t -values in parentheses.

Table 7. *Returns to schooling from IV—male employees 18–45 years (Instrument = 1 if age in 1981 between 2 and 18 years)*

Variable	IV: Reform + father's years of schooling	IV: Reform + father's years of schooling	IV: Reform + father's years of schooling
Years of schooling	0.092 (4.5)	0.054 (1.8)	0.051 (1.7)
Experience	0.012 (0.9)	0.012 (0.9)	0.016 (1.2)
Experience squared	−0.0002 (0.6)	−0.0002 (0.6)	−0.0004 (1.0)
Urban	0.190 (1.9)	0.182 (1.9)	0.186 (1.9)
Father white collar	—	—	0.133 (1.6)
Standardized score	—	0.172 (2.5)	0.157 (2.2)
Constant	5.22 (20.6)	5.58 (16.7)	5.57 (16.2)
Centered R^2	0.188	0.205	0.210
N	535	535	499
<i>First Stage</i>			
Partial R^2	0.291	0.198	0.211
Cragg–Donald Wald F-statistic	108.6	65.3	65.7
[p -Value]	[0.000]	[0.000]	[0.000]
<i>Under-identification test</i>			
H_0 : Matrix under-identified:	219.7	132.4	133.6
Cragg–Donald Wald chi-sq-statistic			
[p -Value]	[0.000]	[0.000]	[0.000]
<i>Over-identification test</i>			
Hansen's J-statistic	1.69	0.72	0.021
[p -Value]	[0.194]	[0.397]	[0.650]
<i>Endogeneity test</i>			
Durbin–Wu–Hausman statistic	0.021	0.112	0.335
[p -Value]	[0.884]	[0.738]	[0.562]

Note: Theoretical sampling weights included. t -Values in parentheses.

Table 8. *Independent effect of parents' years of schooling on earnings—male employees 18–45 years*

Dependent variable: Log-hourly wage	(1)	(2)
Years of schooling	0.058 (4.2)	0.063 (4.7)
Cognitive score	0.140 (3.0)	0.160 (3.6)
Experience	0.007 (0.5)	0.013 (0.9)
Experience squared	−0.0001 (0.3)	−0.0003 (0.7)
Married	0.096 (1.3)	0.057 (0.8)
Urban	0.152 (1.7)	0.154 (1.8)
Father's years	0.010 (1.3)	—
Mother's years	—	0.002 (0.3)
R^2 adj.	0.200	0.207
N	535	564

Note: Theoretical sampling weights included. t -Values in parentheses.

Available online at www.sciencedirect.com



ScienceDirect